Historical Sources, Time and Space, and Historical Interpretation in Scrollytelling

At first, I did not understand why Indigenous feminism was connected to environmental protection and the threat of salmon extinction. Could this be understood as a historical perspective on ecological research? When I read that Cordes and Huff understood relationships with the land in terms of care rather than ownership, I realized that I had taken the concept too literally.[1] This made me think about how our perspectives on data and relationships affect the way we use digital tools. Even with the same tool, we may produce different narratives. Seeing salmon as resources or as kin affects how researchers understand the river's history and which forms of knowledge they include on the webpage.

While reading Zhao Peng, my first thought was that data storytelling actually asks more of the storyteller.[2] Both journalists and historians need to find a clear line of reasoning in scattered materials. Digital tools do not make this task less demanding. Historians must pay particular attention to incomplete sources, sources produced at different times, and uneven preservation. Having more material only creates more possible directions for a narrative. We still need to explain why we choose a particular direction. For example, maps of fish distribution are useful for comparing spatial relationships. However, overlapping distributions alone cannot explain a decline in fish numbers. The years covered by the data, the survey methods, and the gaps in the data all affect how much we can conclude from the comparison.

The StoryMap showed me that visualization can do more than display text and data. It can also turn things in nature into abstract forms that are easier to study.[3] I am also trying to understand "language" as a medium that can communicate meaningful information and can be recorded. This would allow the sounds of a river to become part of research. However, we need to consider what these sounds mean and how they are recorded in each case. "The River's Melody" turns a recording into music. It reminds me to distinguish the sounds recorded from the river from a person's artistic interpretation of them. A recording made today can tell us about the present environment and how people understand the river. It cannot directly recreate the past. When we expand the range of historical sources, we must also explain the historical context of the new material.

I still think that the development of digital technology does not mean that research can be purely digital and separate from "the land." In historical research, I prefer to understand "the land" as physical objects, original archival records, and fieldwork. Working through a screen makes it easier to store, use, and connect materials. It cannot replace questions about where those materials come from and the conditions in which they were produced. I believe that technology cannot formulate research questions. The patterns shown by a tool may suggest questions, but researchers still have to decide why those questions matter and how they relate to a historical context. If we assign precise coordinates to an uncertain place name, or put

records from different periods into the same map layer, the digital display may even seem more certain than the sources themselves.

When considering author-driven and reader-driven approaches, I am more concerned with the relationship between researchers and their audiences. Research writing also needs readers. Without them, how can research be understood, discussed, and recognized? Attracting readers should therefore not be seen as a task for journalism alone. However, attracting attention is different from giving readers room to interpret the material. The StoryMap's section navigation and layer controls make it easy to browse, but readers still move through materials and paths chosen by the authors. As a historian, I would like this interaction to help readers return to the original sources and distinguish facts, inferences, and points of disagreement. Clicking and scrolling alone are not enough to let readers test a researcher's judgments.

I feel that subjectivity appears more directly in how a StoryMap presents its results. Which data are selected, how they are arranged, and which images are used are all connected to the researcher's perspective. Cordes and Huff present salmon, shells, vortices, and music before moving to maps of fish distribution. This helped me understand the relationship between people and the river, but it also made me question the historical link between the two parts. An emotional connection can encourage care, but explaining how the crisis developed still requires attention to policy, land use, and ecological change. I would add links to materials with dates and sources within the existing structure. Readers could then move back and forth to compare them, without mistaking the order of the webpage for an inevitable chain of historical causes and effects.

When using scrollytelling in historical research, my most practical question is: how can I find accurate and suitable historical images? If we decide what images a webpage needs before looking for sources, the demands of presentation may shape the direction of the research. People and places with more surviving photographs may also receive more space in the narrative. When choosing images, I would therefore want to state who took them, their purpose and date, and the conditions in which they were preserved. I would also explain what questions the images can answer. Parts with little surviving material should still be included, with this lack of evidence noted as a limitation of the research. They should not disappear from the narrative simply because there are no images to display. A lack of records does not mean that no historical activity took place.

Even when I find images, I am still unsure whether their details can be reconstructed. To decide whether modern restoration harms an image's historical character and authenticity, we need to distinguish different kinds of treatment. Removing stains and filling in missing parts do not change the evidence in the same way. I would keep the original images for comparison, along with explanations of how they were processed. I would also distinguish reconstructions based on inference from records made at the time. The layout of the text matters too. Images, charts, and text all serve the research. Scrollytelling's visual strengths should not push

disagreements and uncertainties that are difficult to show in images into the background. Pausing to explain the limits of the evidence is also part of historical storytelling.

I also see digital tools as a platform for effective communication between historians and researchers in other disciplines. In river research, historical materials and ecological surveys can each raise questions for the other. Technical specialists can also help identify the limits of different presentation methods. However, this brings me back to a question I had while reading: does digital technology, in some ways, make it harder for marginalized scholars and groups to take part in research? Being able to browse a webpage does not mean being able to take part in selecting and interpreting the materials. Equipment, specialist skills, and funding for maintenance all affect who can have a voice. Low-bandwidth text, explanations of audio, and versions that can be saved should therefore be considered alongside visual design.

I still want to ask whether the subjectivity of a StoryMap weakens its academic credibility. As I currently understand it, the key issues are whether we can trace how choices were made and whether the people represented can take part in the decisions. Following the tribe's advice, the authors removed a location from the map because showing it could have led to looting at an archaeological site. This showed me that the completeness of information and community safety also need to be balanced.[1] When I use tools like this in the future, I will keep records of sources, processing steps, and revisions. I will also involve the relevant groups in decisions about interpretation and the withdrawal of material. Scrollytelling can expand the sources, communication, and forms of presentation available to historical research. However, its power to persuade still needs to rest on historical interpretations that can be checked and questioned.

References

[1] Cordes, Ashley, and Micah Huff. “Decolonial Feminist Storying on the Coquille River: A Digital Humanities Approach to Human and Non-human Communication and Prevention of the Fall Chinook Salmon Extinction.” Feminist Formations 35, no. 1 (2023): 73–96. This reflection mainly refers to pp. 76–80 and 84–90. DOI: 10.1353/ff.2023.a902067

[2] Zhao Peng. “Telling a Story with Graphs: How to Apply Narrative Visualization Strategies and Visualization Techniques in Journalistic Storytelling.” In The Palgrave Handbook of Global Digital Journalism, edited by Bruce Mutsvairo and Kristin Skare Orgeret, 283–306. Palgrave Macmillan, 2025. This reflection mainly refers to pp. 288–292. DOI: 10.1007/978-3-031-59379-6_18

[3] “Storying on the Coquille River.” ArcGIS StoryMaps, 2021. Accessed October 19, 2026. StoryMap webpage

Statement

The translation was completed with the assistance of the AI tool ChatGPT. The original composition was written in Chinese.